



Linux Practical Assignment: Basic Commands and File System

Course: Introduction to Linux Operating System

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Assignment Title: Exploring Linux Commands and File System Navigation

Objective

This assignment aims to help students understand the Linux file system, differentiate between relative and absolute paths, and effectively use basic file management and navigation commands.

Part A: Theoretical Section

1. Linux File System Overview

- Explain the structure of the Linux file system starting from the root (/).
- Describe the purpose of the following directories:
 - /home
 - /etc
 - /bin
 - /var
 - /usr

- /tmp
- Differentiate between absolute and relative paths.
 - Provide examples of each.

2. Commands Overview

Write short notes on the following commands:

- ls
- mkdir
- pwd
- nano
- rm
- touch

Include two (2) common options for each command and explain their functions.

Part B: Practical Section

Scenario: Working in Your Home Directory

You are a junior system administrator organizing files for a new project named `cyber_project`.

Follow the steps below and write the commands you used.

Task 1: Directory Management

1. Navigate to your home directory.
2. Create a directory named `cyber_project`.
3. Inside `cyber_project`, create three subdirectories:
 - `reports`
 - `scripts`
 - `configs`
4. Use one command to create all three directories at once.

Task 2: File Creation and Editing

1. Inside the `reports` directory, create an empty file named `summary.txt`.
Use the `nano` editor to open and type the following text inside `summary.txt`:

Project: Cyber Security Lab

Author: <Your Name>

Date: <Today's Date>

2. Save and exit the file: Inside the `scripts` directory, create another file named `setup.sh`.
Add this line in it using nano:
`#!/bin/bash`
`echo "Setting up the cyber project environment..."`

Task 3: File and Directory Listing

1. List all files and directories inside `cyber_project` in long format.
2. List hidden files if any.
3. Display your current directory path.

Task 4: Paths

1. Using relative path, open the `summary.txt` file from inside `cyber_project`.
2. Using absolute path, open the `setup.sh` file.

Task 5: Deleting Files and Directories

1. Create a new empty file called `test.txt` inside the `configs` directory.
2. Delete the file using the `rm` command.
3. Delete the entire `configs` directory (make sure it's empty first).

Task 6: Bonus Challenge

1. Use `mkdir -p` to create nested directories in one command:
`mkdir -p cyber_project/backups/2025/October`
2. Use `ls -R` to display all files and directories recursively inside `cyber_project`.

Part C: Submission

- Submit a document (Word or PDF) containing:
 - Screenshots of each task.
 - The commands you used.
 - Short explanations (1–2 sentences) of what each command does.
 - Your theory answers.

Marking Scheme

Section	Description	Marks
Theory (File System + Paths)	Explanation and examples	20
Basic Commands	Notes and options	20
Practical Tasks	Commands and screenshots	50
Bonus Challenge	Optional tasks	10
Total		100